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Spring REST Using Spring Boot 3

1) Spring web project using maven

SpringLearnApplication.java

```
package com.cognizant.springlearn;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication

public class SpringLearnApplication {

    public static void main(String[] args) {

        System.out.println("Application Starting...");

        SpringApplication.run(

            SpringLearnApplication.class,

            args);

        System.out.println("Application Started...");

    }

}
```

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
```

```
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
    https://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<parent>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-parent</artifactId>
    <version>3.5.0</version>
    <relativePath/>
</parent>
<groupId>com.cognizant</groupId>
<artifactId>spring-learn</artifactId>
<version>0.0.1-SNAPSHOT</version>
<name>spring-learn</name>
<properties>
    <java.version>17</java.version>
</properties>
<dependencies>
    <dependency>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-web</artifactId>
    </dependency>
    <dependency>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-devtools</artifactId>
        <scope>runtime</scope>
```

```
        <optional>true</optional>
    </dependency>
    <dependency>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-test</artifactId>
        <scope>test</scope>
    </dependency>
</dependencies>
<build>
    <plugins>
        <plugin>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-maven-plugin</artifactId>
        </plugin>
    </plugins>
</build>
</project>
```

application.properties

```
spring.application.name=spring-learn
server.port=8080
```

SpringLearnApplicationTest.java

```
package com.cognizant.springlearn;
```

```

import org.junit.jupiter.api.Test;
import org.springframework.boot.test.context.SpringBootTest;
@SpringBootTest
class SpringLearnApplicationTests {

    @Test
    void contextLoads() {
    }

}

```

2) Spring Core – Load Country from Spring Configuration XML

country.xml

```

<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
       xsi:schemaLocation="
           http://www.springframework.org/schema/beans
           https://www.springframework.org/schema/beans/spring-
beans.xsd">
    <bean id="country"
          class="com.cognizant.springlearn.Country">
        <property name="code" value="IN"/>

        <property name="name" value="India"/>
    </bean>

```

</beans>

Country.java

```
package com.cognizant.springlearn;

public class Country {

    private String code;

    private String name;

    public Country() {

        System.out.println("Inside Country Constructor.");

    }

    public String getCode() {

        System.out.println("Inside getCode()");

        return code;

    }

    public void setCode(String code) {

        System.out.println("Inside setCode()");

        this.code = code;

    }

    public String getName() {

        System.out.println("Inside getName()");

        return name;

    }

    public void setName(String name) {

        System.out.println("Inside setName()");

        this.name = name;

    }

}
```

```

    }
    @Override
    public String toString() {
        return "Country [code=" + code + ", name=" + name + "];"
    }
}

```

SpringLearnApplication.java

```

package com.cognizant.springlearn;

import org.springframework.context.ApplicationContext;
import
org.springframework.context.support.ClassPathXmlApplicationContext;

public class SpringLearnApplication {
    public static void main(String[] args) {
        displayCountry();
    }

    public static void displayCountry() {
        ApplicationContext context =
            new ClassPathXmlApplicationContext("country.xml");
        Country country =
            context.getBean("country", Country.class);
        System.out.println("Country : " + country);
    }
}

```

3) Hello World RESTful Web Service

HelloController.java

```
package com.cognizant.springlearn.controller;

import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RestController;

@RestController
public class HelloController {

    private static final Logger LOGGER =
        LoggerFactory.getLogger(HelloController.class);

    @GetMapping("/hello")
    public String sayHello() {

        LOGGER.info("START");

        String message = "Hello World!!";

        LOGGER.info("END");

        return message;

    }

}
```

application.properties

```
server.port=8083

spring.application.name=spring-learn
```

SpringLearnApplication.java

```
package com.cognizant.springlearn;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
@SpringBootApplication
public class SpringLearnApplication {

    public static void main(String[] args) {

        SpringApplication.run(SpringLearnApplication.class, args);

    }

}
```

pom.xml Dependency

```
<dependency>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-web</artifactId>
</dependency>
```


4) REST - Country Web Service

Country.java

```
package com.cognizant.springlearn;

public class Country {
    private String code;
    private String name;
    public Country() {
    }
    public String getCode() {
        return code;
    }
    public void setCode(String code) {
        this.code = code;
    }
    public String getName() {
        return name;
    }
    public void setName(String name) {
        this.name = name;
    }
}
```

country.xml

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<beans xmlns="http://www.springframework.org/schema/beans"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="
    http://www.springframework.org/schema/beans
    https://www.springframework.org/schema/beans/spring-beans.xsd">
  <bean id="country"
    class="com.cognizant.springlearn.Country">
    <property name="code" value="IN"/>
    <property name="name" value="India"/>
  </bean>
</beans>
```

CountryController.java

```
package com.cognizant.springlearn.controller;

import com.cognizant.springlearn.Country;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.bind.annotation.RestController;

@RestController

public class CountryController {

    @RequestMapping("/country")

    public Country getCountryIndia() {
```

```

ApplicationContext context =
    new ClassPathXmlApplicationContext("country.xml");
Country country =
    context.getBean("country", Country.class);
return country;
}
}

```

5) REST - Get country based on country code

Country.java

```

package com.cognizant.springlearn;

public class Country {
    private String code;
    private String name;
    public Country() {
    }
    public String getCode() {
        return code;
    }
    public void setCode(String code) {
        this.code = code;
    }
    public String getName() {
        return name;
    }
}

```

```
public void setName(String name) {  
    this.name = name;  
}  
}
```

country.xml

```
<?xml version="1.0" encoding="UTF-8"?>  
<beans xmlns="http://www.springframework.org/schema/beans"  
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"  
    xsi:schemaLocation="  
        http://www.springframework.org/schema/beans  
        https://www.springframework.org/schema/beans/spring-  
beans.xsd">  
    <bean id="countryList"  
        class="java.util.ArrayList">  
        <constructor-arg>  
            <list>  
                <bean class="com.cognizant.springlearn.Country">  
                    <property name="code" value="US"/>  
                    <property name="name" value="United States"/>  
                </bean>  
                <bean class="com.cognizant.springlearn.Country">  
                    <property name="code" value="DE"/>  
                    <property name="name" value="Germany"/>  
                </bean>
```

```

        <bean class="com.cognizant.springlearn.Country">
            <property name="code" value="IN"/>
            <property name="name" value="India"/>
        </bean>
        <bean class="com.cognizant.springlearn.Country">
            <property name="code" value="JP"/>
            <property name="name" value="Japan"/>
        </bean>
    </list>
</constructor-arg>
</bean>
</beans>

```

CountryService.java

```

package com.cognizant.springlearn.service;

import java.util.List;

import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import org.springframework.stereotype.Service;

import com.cognizant.springlearn.Country;

@Service
public class CountryService {

    public Country getCountry(String code) {

```

```

    ApplicationContext context =
        new ClassPathXmlApplicationContext("country.xml");
    List<Country> countries =
        (List<Country>) context.getBean("countryList");
    return countries.stream()
        .filter(c -> c.getCode().equalsIgnoreCase(code))
        .findFirst()
        .orElse(null);
}
}

```

CountryController.java

```

package com.cognizant.springlearn.controller;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PathVariable;
import org.springframework.web.bind.annotation.RestController;
import com.cognizant.springlearn.Country;
import com.cognizant.springlearn.service.CountryService;
@RestController
public class CountryController {
    @Autowired
    private CountryService countryService;
    @GetMapping("/countries/{code}")
    public Country getCountry(

```

```
        @PathVariable String code) {  
            return countryService.getCountry(code);  
        }  
    }  
}
```

6) Create authentication service that returns JWT

AuthenticationResponse.java

```
package com.cognizant.springlearn.model;  
  
public class AuthenticationResponse {  
    private String token;  
  
    public AuthenticationResponse(String token) {  
        this.token = token;  
    }  
  
    public String getToken() {  
        return token;  
    }  
  
    public void setToken(String token) {  
        this.token = token;  
    }  
}
```

AuthenticationController.java

```
package com.cognizant.springlearn.controller;
```

```
import java.nio.charset.StandardCharsets;
import java.util.Base64;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RequestHeader;
import org.springframework.web.bind.annotation.RestController;
import com.cognizant.springlearn.model.AuthenticationResponse;

@RestController
public class AuthenticationController {

    @GetMapping("/authenticate")
    public AuthenticationResponse authenticate(
        @RequestHeader("Authorization") String authHeader) {
        String base64Credentials =
            authHeader.substring("Basic ".length());
        byte[] decodedBytes =
            Base64.getDecoder()
                .decode(base64Credentials);
        String credentials =
            new String(decodedBytes,
                StandardCharsets.UTF_8);
        String[] values =
            credentials.split(":", 2);
        String username = values[0];
        String token =
            "eyJhbGciOiJIUzI1NiJ9.eyJzdWUiOi"
            + username
```



```
        + ".sampletoken";  
    return new AuthenticationResponse(token);  
}  
}
```

SecurityConfig.java

```
package com.cognizant.springlearn.config;  
  
import org.springframework.context.annotation.Bean;  
import org.springframework.context.annotation.Configuration;  
import org.springframework.security.config.Customizer;  
  
import  
org.springframework.security.config.annotation.web.builders.HttpSecurity;  
import org.springframework.security.core.userdetails.User;  
import org.springframework.security.core.userdetails.UserDetails;  
  
import  
org.springframework.security.provisioning.InMemoryUserDetailsManager;  
import org.springframework.security.web.SecurityFilterChain;  
  
@Configuration  
public class SecurityConfig {  
  
    @Bean  
  
    public InMemoryUserDetailsManager userDetailsService() {  
        UserDetails user =  
            User.withDefaultPasswordEncoder()  
                .username("user")  
                .password("pwd")  
                .roles("USER")
```

```

        .build();

        return new InMemoryUserDetailsManager(user);
    }

    @Bean
    public SecurityFilterChain filterChain(
        HttpSecurity http) throws Exception {
        http
            .authorizeHttpRequests(auth ->
                auth.anyRequest().authenticated())
            .httpBasic(Customizer.withDefaults());
        return http.build();
    }
}

```

pom.xml Dependency

```

<dependency>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-security</artifactId>
</dependency>

```